Scalable Poisson Mixture Learning from Probability-Generating Function Sketches

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Abstract

Finite mixtures of Poisson distributions are a standard model for heterogeneous count data, but likelihood-based inference such as Expectation–Maximization is iterative and scales with the dataset size. We study a compressed alternative: the dataset is summarized once by sampling its empirical probability-generating function (PGF) along complex lines, yielding a fixed-size sketch. For a finite mixed-Poisson model this sketch is a multidimensional sum of exponentials, which a joint ESPRIT procedure turns directly into the mixture rate vectors and weights, without any combinatorial cross-coordinate pairing. Damping the query points keeps the sketch numerically stable on overdispersed data. Because the sketch size is independent of the number of observations n, both storage and recovery cost are independent of n; our experiments confirm that the recovery stays flat as full-data EM grows linearly, while the sketch is orders of magnitude smaller than the raw data.

CCS Concepts

• **Information systems** → **Data mining**; • **Mathematics of computing** → *Probabilistic algorithms*; • **Computing methodologies** → *Mixture models*.

Keywords

data sketching, Poisson mixtures, count data, probability generating functions, ESPRIT, joint diagonalization

1 Introduction

Many datasets consist of nonnegative integer-valued observations: numbers of events, interactions, messages, visits, reads, or molecular counts. Finite mixtures of Poisson distributions provide a simple and interpretable model for such data. They describe heterogeneous populations through a finite number of latent components, each associated with a Poisson rate vector and a mixture weight. Estimating these parameters is a classical latent-variable inference problem, most commonly addressed through maximum-likelihood methods such as Expectation–Maximization (EM) [4, 7]. Although effective, EM-type procedures remain iterative, can be sensitive to initialization, and require repeated passes over the data or over sufficient statistics whose size grows with the model and data dimension.

Sketching offers an alternative route for large-scale inference. Instead of fitting the model directly from all observations, the dataset is first compressed into a fixed-size collection of nonlinear measurements, called a sketch, and estimation is performed from this compact representation [3, 8]. For count data, a natural choice is the probability generating function (PGF). Indeed, if X is a Poisson random variable with rate λ, then its PGF is given by

$$G_X(t) = \mathbb{E}[t^X] = \exp(\lambda(t-1)).$$

Consequently, the PGF of a finite Poisson mixture is a weighted sum of exponentials. This algebraic structure suggests that mixture estimation from empirical PGF samples can be viewed as a harmonic retrieval problem, opening the door to spectral methods such as ESPRIT [9].

In this paper, we study a multidimensional sketching approach for Poisson mixture estimation based on *joint ESPRIT*. The method evaluates the empirical PGF along random-base complex lines, stacks the resulting multi-snapshot measurements, and estimates a global signal subspace shared across directions. The Poisson rate vectors are then recovered through jointly diagonalized rotational invariance matrices, while the mixture weights are estimated by solving a linear inverse problem. Compared with coordinate-wise spectral recovery, the joint formulation is designed to avoid a combinatorial alignment step between independently estimated marginal components.

Contributions. The contributions of this work are as follows:

- We cast finite Poisson mixture estimation from empirical PGF sketches as a multidimensional harmonic retrieval problem, and introduce a *damped* query construction that keeps the sketch numerically stable while preserving its exact sum-of-exponentials structure.
- We adapt a random-base joint ESPRIT pipeline that recovers multidimensional Poisson rate vectors from a fixed-size sketch without coordinate-wise pairing, followed by a constrained recovery of rates and weights and a refinement that minimizes the exact PGF loss.
- We show empirically that, because the sketch size is fixed, the storage and recovery cost of the method are independent of the number of observations, in contrast to full-data EM.

2 Model and PGF Sketch

Let $X \in \mathbb{N}^d$ denote an observed count vector. We consider the finite product-Poisson mixture

$$Z \sim \sum_{k=1}^{K} \pi_k \delta_{\lambda_k}, \qquad X \mid Z = \lambda_k \sim \bigotimes_{q=1}^{d} \mathrm{Poisson}(\lambda_{qk}), \tag{1}$$

where $\lambda_k \in \mathbb{R}_+^d$, $\pi_k \geq 0$, and $\sum_{k=1}^K \pi_k = 1$. We write $\Lambda = [\lambda_1, \ldots, \lambda_K] \in \mathbb{R}_+^{d\times K}$ and $\pi = (\pi_1, \ldots, \pi_K)^\top$.

The probability generating function of X is

$$G_X(t) = \mathbb{E}\left[\prod_{q=1}^{d} t_q^{X_q}\right] = \sum_{k=1}^{K} \pi_k \exp(\langle \lambda_k, t - \mathbf{1}\rangle), \qquad t \in \mathbb{C}^d. \tag{2}$$

Given independent samples $x^{(1)}, \ldots, x^{(n)}$, we use the empirical PGF

$$\widehat{G}_X(t) = \frac{1}{n}\sum_{i=1}^{n}\prod_{q=1}^{d} t_q^{x_q^{(i)}} = \frac{1}{n}\sum_{i=1}^{n} \exp\Big(\langle x^{(i)}, \log(t)\rangle\Big), \tag{3}$$

where the logarithm is applied coordinate-wise. Since the exponents $x_q^{(i)}$ are integers, this coincides with the polynomial definition of the PGF whenever the coordinates of t are nonzero.

Random-base line sampling. We choose M probing directions

$$u_1, \dots, u_M \in \mathbb{R}^d, \quad U = [u_1^\top; \dots; u_M^\top] \in \mathbb{R}^{M \times d},$$

with $\operatorname{rank}(U) = d$. We also draw S random base points $p_1, \dots, p_S \in \mathbb{R}^d$, shared across all directions, and sample the lines

$$\xi_{m,s,\ell} = p_s + \ell u_m, \quad m = 1, \dots, M, \quad s = 1, \dots, S, \quad \ell = 0, \dots, N_m - 1. \tag{4}$$

Figure 1 illustrates this random-base line sampling in two dimensions. The PGF is evaluated at the *damped* complex query points

$$t_{m,s,\ell} = r\,\mathbf{1} + \mathrm{i}\,\delta_{\text{eff}}\,\xi_{m,s,\ell}, \qquad |t_{m,s,\ell,q}| \le 1 \ \ \forall q, \tag{5}$$

where $r \in (0, 1]$ is a real damping radius and $\delta_{\text{eff}} > 0$ an effective sketching scale (the marginal choice $t = \mathbf{1} + \mathrm{i}\Delta\xi$ is the undamped case $r = 1$, $\delta_{\text{eff}} = \Delta$). The radius must be chosen with care. For a single observation, the empirical PGF term has modulus $\prod_q |t_q|^{x_q^{(i)}}$: with $|t_q| > 1$ it grows geometrically in the counts, so $\widehat{G}_X$ becomes high-variance on overdispersed data, while a very small r pulls every query point toward $\mathbf{1}$ and washes out the exponential structure. A good radius therefore balances this damping bias against the $O(n^{-1/2})$ sampling fluctuation of $\widehat{G}_X$; we use a simple data-driven rule, detailed in the released code, that keeps every query point inside the unit disk.

At the population level, the damped queries still give an *exact* sum of exponentials. Since $t_{m,s,\ell} - \mathbf{1} = (r-1)\mathbf{1} + \mathrm{i}\delta_{\text{eff}}\,\xi_{m,s,\ell}$,

$$\begin{aligned} G_X(t_{m,s,\ell}) &= \sum_{k=1}^{K} \pi_k \exp\Big((r-1)\langle \lambda_k, \mathbf{1}\rangle + \mathrm{i}\delta_{\text{eff}}\langle \lambda_k, p_s + \ell u_m\rangle\Big) \\ &= \sum_{k=1}^{K} c_{k,s}\,\rho_{m,k}^{\ell}, \end{aligned} \tag{6}$$

with

$$c_{k,s} = \pi_k \underbrace{e^{(r-1)\langle \lambda_k, \mathbf{1}\rangle}}_{\text{real damping}} e^{\mathrm{i}\delta_{\text{eff}}\langle \lambda_k, p_s\rangle}, \ \ \rho_{m,k} = e^{\mathrm{i}\phi_{m,k}}, \ \ \phi_{m,k} = \delta_{\text{eff}}\langle \lambda_k, u_m\rangle. \tag{7}$$

The key structural point is that $|\rho_{m,k}| = 1$: the snapshot index ℓ enters *only* through the imaginary increment $\mathrm{i}\delta_{\text{eff}}\,\ell\,u_m$, while the real damping $(r-1)\langle \lambda_k, \mathbf{1}\rangle$ is constant along each line and is absorbed into the coefficient $c_{k,s}$. Consequently damping leaves the ESPRIT nodes on the unit circle and changes neither the subspace nor the diagonalization algebra below; it only rescales the recovery scale $\Delta \to \delta_{\text{eff}}$. The PGF sketch is thus not a linearization: for the query family (5) the exponential model is exact, up to empirical PGF noise.

The empirical observations are

$$z_{m,s}[\ell] := \widehat{G}_X(t_{m,s,\ell}) = \sum_{k=1}^{K} c_{k,s}\rho_{m,k}^{\ell} + e_{m,s}[\ell], \tag{8}$$

where $e_{m,s}[\ell] = \widehat{G}_X(t_{m,s,\ell}) - G_X(t_{m,s,\ell})$ is the finite-sample sketching error.

Multi-snapshot matrix model. For each direction m, stack the S line measurements into

$$Z_m = \begin{bmatrix} z_{m,1}[0] & \cdots & z_{m,S}[0] \\ \vdots & & \vdots \\ z_{m,1}[N_m - 1] & \cdots & z_{m,S}[N_m - 1] \end{bmatrix} \in \mathbb{C}^{N_m \times S}.$$

Define

$$v_m(\phi) = [1, e^{\mathrm{i}\phi}, \dots, e^{\mathrm{i}(N_m-1)\phi}]^\top, \qquad V_m = [v_m(\phi_{m,1}), \dots, v_m(\phi_{m,K})],$$

and

$$C = (c_{k,s})_{k,s} \in \mathbb{C}^{K \times S}.$$

Then

$$Z_m = V_m C + E_m. \tag{9}$$

Equivalently, stacking all directions gives the global multi-directional model

$$Z_{\text{glob}} = \begin{bmatrix} Z_1 \\ \vdots \\ Z_M \end{bmatrix} = \underbrace{\begin{bmatrix} V_1 \\ \vdots \\ V_M \end{bmatrix}}_{V_{\text{glob}}} C + E_{\text{glob}}. \tag{10}$$

The use of shared random base points is important: it gives a common right factor C across all directions, and therefore a common component basis for the joint ESPRIT step.

We assume throughout that $N_m \ge K + 1$, $S \ge K$, that the projected phases $\phi_{m,k}$ are distinct modulo 2π for each direction used in ESPRIT and that C has full row rank. For random base points, this rank condition holds generically provided the components are distinct and the number of snapshots is large enough.

3 Joint ESPRIT Recovery

The recovery follows the multi-directional ESPRIT structure suggested by (10). The goal is to estimate the projected phases $\phi_{m,k} = \delta_{\text{eff}}\langle \lambda_k, u_m\rangle$ with a consistent component index k across all directions.

Global subspace estimation. We compute a rank-K left signal subspace from the global sketch matrix (10). For instance, let

$$\widehat{R} = \frac{1}{S} Z_{\text{glob}} Z_{\text{glob}}^{*} \tag{11}$$

and let $\widehat{U}_s \in \mathbb{C}^{(\sum_m N_m) \times K}$ contain the K dominant eigenvectors of $\widehat{R}$. In the noiseless case, there exists an invertible matrix $T_0 \in \mathbb{C}^{K \times K}$ such that

$$\widehat{U}_s = V_{\text{glob}} T_0. \tag{12}$$

Let $\widehat{U}_{s,m} \in \mathbb{C}^{N_m \times K}$ denote the block of $\widehat{U}_s$ corresponding to direction m.

Rotational invariance per direction. Let $J_m^{(1)} \in \mathbb{R}^{(N_m-1) \times N_m}$ remove the last row and $J_m^{(2)} \in \mathbb{R}^{(N_m-1) \times N_m}$ remove the first row. In the noiseless model,

$$J_m^{(2)} V_m = J_m^{(1)} V_m \operatorname{diag}(\rho_{m,1}, \dots, \rho_{m,K}). \tag{13}$$

Since $\widehat{U}_{s,m}$ spans the same column space as V_m, we estimate a rotational invariance matrix by weighted least squares:

$$\widehat{\Phi}_m \in \arg\min_{\Phi \in \mathbb{C}^{K \times K}} \left\| W_m \left(J_m^{(2)} \widehat{U}_{s,m} - J_m^{(1)} \widehat{U}_{s,m} \Phi \right) \right\|_F^2. \tag{14}$$

Here W_m is an optional row-weighting matrix. In the simplest implementation $W_m = I$. In practice, W_m can down-weight PGF evaluations with high empirical variance or unstable numerical magnitude. This plays the same stabilizing role as row weighting in PSF-aware ESPRIT, without requiring any inversion of small factors.

In the noiseless case,

$$\Phi_m = T_0^{-1}\,\mathrm{diag}(\rho_{m,1},\dots,\rho_{m,K})T_0, \tag{15}$$

so the matrices $\Phi_1,\dots,\Phi_M$ are simultaneously diagonalizable and share the same eigenvectors.

Similarity joint diagonalization and component alignment. By (15) the matrices $\widehat{\Phi}_1,\dots,\widehat{\Phi}_M$ share a common eigenbasis, but this basis is the columns of T_0, an *arbitrary invertible* matrix, not an orthogonal one. A unitary joint-diagonalization sweep, which restricts the change of basis to be orthogonal, is therefore not guaranteed to diagonalize them. We instead estimate the common change of basis by a *non-unitary* (similarity) joint diagonalization:

$$\widehat{T} \in \arg\min_{T\in \mathrm{GL}(K)} \sum_{m=1}^{M} \left\| \mathrm{offdiag}\left(T\widehat{\Phi}_m T^{-1}\right)\right\|_F^2. \tag{16}$$

Unlike a unitary Jacobi sweep [2], which constrains the change of basis to be orthogonal, this non-unitary scheme allows the genuinely non-orthogonal eigenbasis T_0 of the ESPRIT matrices. The jointly aligned node estimates are then obtained as

$$\widehat{\rho}_{m,k} = \left(\widehat{T}\widehat{\Phi}_m\widehat{T}^{-1}\right)_{kk}, \qquad \widehat{\phi}_{m,k} = \arg(\widehat{\rho}_{m,k}). \tag{17}$$

The common diagonalizer fixes the component ordering across directions, avoiding a separate combinatorial pairing step.

Rate reconstruction. The aligned phases satisfy $\widehat{\phi}_{m,k} \approx \delta_{\mathrm{eff}}\langle\lambda_k, u_m\rangle \pmod{2\pi}$. The damped scale is chosen so that the phases do not wrap; dropping the modulus, we form the directional projections $g_{m,k} = \widehat{\phi}_{m,k}/\delta_{\mathrm{eff}} \approx \langle\lambda_k, u_m\rangle$ and recover each rate by non-negative least squares,

$$\widehat{\lambda}_k = \arg\min_{\lambda\in\mathbb{R}_+^d} \|U\lambda - g_k\|_2^2, \qquad g_k = [g_{1,k},\dots,g_{M,k}]^\top, \tag{18}$$

which imposes nonnegative rates. The condition $\mathrm{rank}(U) = d$ makes the rates identifiable from their directional projections.

Mixture-weight estimation. Once $\widehat{\Lambda}$ has been recovered, the mixture weights are estimated from all available PGF samples by a global simplex-constrained least-squares fit:

$$\widehat{\pi} \in \arg\min_{p\in\Delta_K} \sum_{m=1}^{M}\sum_{s=1}^{S}\sum_{\ell=0}^{N_m-1} \alpha_{m,s,\ell}\left|\widehat{G}_X(t_{m,s,\ell}) - \sum_{k=1}^{K} p_k\, e^{\langle\widehat{\lambda}_k,\, t_{m,s,\ell}-\mathbf{1}\rangle}\right|^2, \tag{19}$$

where the model term reuses the damped query points, $\langle\widehat{\lambda}_k, t_{m,s,\ell} - \mathbf{1}\rangle = (r-1)\langle\widehat{\lambda}_k, \mathbf{1}\rangle + \mathrm{i}\delta_{\mathrm{eff}}\langle\widehat{\lambda}_k, p_s + \ell u_m\rangle$, and the feasible set is the probability simplex

$$\Delta_K = \{p \in \mathbb{R}_+^K : \mathbf{1}^\top p = 1\}.$$

This is a convex quadratic program in p, which we solve with a standard simplex-constrained least-squares solver. The weights $\alpha_{m,s,\ell} \geq 0$ may be set to one or used to down-weight numerically unstable query points.

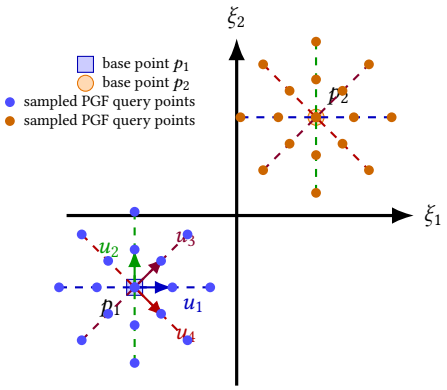


Figure 1: Two-dimensional illustration of the random-base line sampling strategy. Two base points p_1 and p_2 are chosen randomly. Around each base point, four directions are probed, and five query locations are sampled along each direction. For visual clarity, the line samples are drawn symmetrically around the base point; the algorithm only requires consecutive samples along each direction.

Refinement of the exact PGF loss. The recovery above is algebraic and does not directly minimize a fit criterion. We therefore refine $(\widehat{\Lambda}, \widehat{\pi})$ by minimizing the exact PGF least-squares loss over all $J = \sum_m N_m S$ query points $t_1,\dots,t_J$ (we write $y_j = \widehat{G}_X(t_j)$):

$$F(\Lambda,\pi) = \frac{1}{2J}\sum_{j=1}^{J}\left|\sum_{k=1}^{K}\pi_k\, e^{\langle\lambda_k,\, t_j-\mathbf{1}\rangle} - y_j\right|^2. \tag{20}$$

The same objective underlies sketched-mixture learning by preconditioned gradient descent (PGD) [6]; we minimize it instead with a limited-memory quasi-Newton method (L-BFGS), using a closed-form analytic gradient and a log/softmax reparametrization that keeps the rates positive and the weights on the simplex. Initialized at the spectral estimate, it converges in a few iterations and yields an accurate fit directly after the spectral step.

4 Experiments

The experiments evaluate whether PGF–JESPRIT provides an accurate and scalable estimator for finite Poisson mixtures. We explicitly distinguish algorithmic comparisons from model-level comparisons. In the algorithmic setting, all methods target the same finite product-Poisson mixture model in (1). In the model-level setting, we also report results for alternative count-data models, such as Poisson–Gamma variants, but these are not interpreted as direct algorithmic baselines for PGF–JESPRIT.

Synthetic mixtures. We generate n independent samples from the finite Poisson mixture model in (1). The protocol varies the number of components K, the dimension d, the sample size n, the separation between rate vectors, and the dynamic range of the Poisson rates. We consider three representative regimes:

- *Separated components*, where the rate vectors are well separated and parameter recovery should be feasible;

- *Overlapping components*, where several rate vectors are close and the inverse problem becomes harder;
- *Under-specified fitting*, where the fitted number of components is smaller than the true one, to test robustness under model mismatch.

For the scalability study reported below we fix $K = 3$ components in $d = 3$ dimensions, use a sketch of size $M = S = L = 9$ (so $J = 729$), and sweep n from 10^3 to 3×10^5 across the separated and overlapping regimes.

Compared methods. For algorithmic comparison, the main comparator is EM for the finite product-Poisson mixture model in (1). This comparison keeps the probabilistic model fixed and contrasts likelihood-based inference with the proposed PGF-sketching and spectral-recovery pipeline. We report the proposed estimator both directly after spectral recovery (*PGF–ℐESPRIT*) and after the analytic-gradient L-BFGS refinement of the exact PGF loss (20) (*PGF–ℐESPRIT + L-BFGS*). We also report runtime and storage to assess the benefit of replacing repeated likelihood-based fitting by a fixed-size sketch followed by spectral recovery. A coordinate-wise PGF–ESPRIT variant may be included only as an ablation of the proposed spectral pipeline, in order to quantify the effect of the global subspace estimation and joint diagonalization.

When Gamma-regularized or Poisson–Gamma EM variants are included, they are reported separately as model-level comparators. These variants can be useful on overdispersed real count data, but they do not isolate the effect of the inference algorithm because they rely on different probabilistic assumptions or regularization criteria.

Metrics. On synthetic data, where the ground truth is known, we report the relative error on the rate matrix and on the mixture weights after optimal component matching. For both synthetic and real data, we report the PGF reconstruction error, the Hellinger distance between empirical and regenerated marginal distributions, a stabilized chi-square distance, runtime, and the stored representation size ($2J$ sketch scalars vs. nd raw scalars). The distributional metrics are computed per coordinate and then averaged over dimensions.

Sketch parameters. The estimator is controlled by the number of sampling directions M, the number of shared base points (snapshots) S, the number of PGF samples per direction $L = N_m$, and the nominal sampling scale Δ, from which the damping fixes the effective scale δ_{eff} and radius r so that every query stays in the unit disk. The sketch is then $J = MSL$ complex PGF evaluations, stored as $2J$ real scalars, independently of n. Unless otherwise stated these are fixed across trials. We also report a short sensitivity study in Δ, which trades node conditioning against phase wrapping: too small a Δ makes the exponential nodes hard to separate, while too large a Δ pushes the phases $\delta_{\text{eff}}\langle \lambda_k, u_m \rangle$ toward $\pm\pi$.

Real count data. We include small real-data checks on the Affairs count dataset [5] and selected high-count genes from the PBMC3k single-cell RNA-seq benchmark [1]. These experiments are not intended to claim that finite Poisson mixtures are universally valid generative models for these datasets. Rather, they test whether the fitted mixtures reproduce realistic count histograms, including sparsity, imbalance, and overdispersion. On these datasets, Poisson–Gamma variants may be reported as model-level comparators, separately from the finite-Poisson algorithmic comparison.

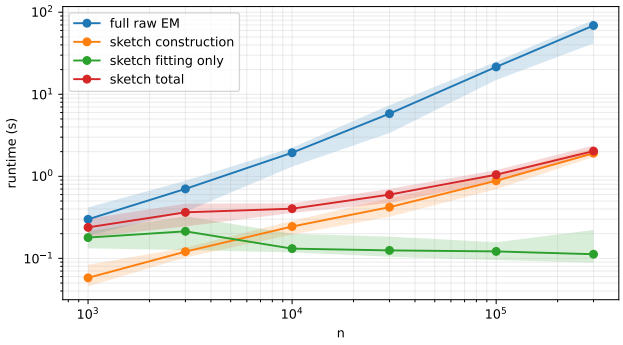


Figure 2: Runtime vs. number of samples n at a fixed sketch size ($M = S = L = 9$, $J = 729$, $K = d = 3$); medians over seeds and regimes with interquartile bands, log–log.

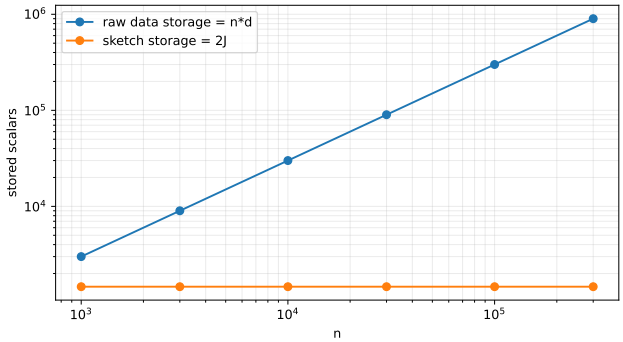


Figure 3: Stored scalars vs. number of samples n at the same fixed sketch size as Fig. 2, log–log.

Scalability and storage. Figures 2 and 3 quantify the central computational claim: at a fixed sketch size, both the storage and the recovery cost are independent of the number of observations n. Figure 2 splits the runtime into four curves: full-data EM; the single streaming pass that builds the sketch (*sketch construction*); the spectral fitting on the sketch (*sketch fitting*: global SVD, similarity diagonalization, rate and weight recovery, and the L-BFGS refinement); and their sum (*sketch total*). The fitting curve is essentially flat in n, since it acts only on the fixed-size sketch, whereas EM grows linearly in n per iteration; the only data-dependent cost is the sketch construction. As a result the sketch pipeline overtakes EM beyond $n \approx 10^4$ and is about an order of magnitude faster at the largest sample size we test ($\approx 35\times$ at $n = 3 \times 10^5$). Figure 3 shows the matching storage picture: the sketch is a fixed $2J$ real scalars while the raw data grows as nd, so at $n = 3 \times 10^5$ with $d = 3$ the sketch ($\approx 1.5 \times 10^3$ scalars) is more than two orders of magnitude smaller than the raw data ($\approx 9 \times 10^5$).

Discussion and limitations. We expect PGF–JESPRIT to be most effective when the mixture has a moderate number of components and the rate vectors are sufficiently separated, so that the empirical PGF has a clear exponential structure and the joint subspace recovers coherent components. Several limitations remain. First, the nominal scale Δ still matters because it affects both node conditioning and phase wrapping; the damping fixes the *magnitude* instability of the undamped queries (it forces $|t| \leq 1$ and matches the bias to the $n^{-1/2}$ sampling fluctuation), but it does not remove the conditioning/wrapping trade-off, which is why we report a Δ sensitivity study. Second, the spectral recovery is algebraic and does not directly optimize a fit criterion; the L-BFGS refinement of (20) mitigates this by minimizing the exact PGF loss after recovery, but in highly overlapping regimes a good PGF fit may still trail EM on histogram-based metrics. Finally, comparisons with Poisson–Gamma variants should be read as model-level comparisons, since differences may reflect both the inference procedure and the probabilistic assumptions.

5 Conclusion

We presented a scalable framework for mixed Poisson inference from empirical PGF sketches using joint ESPRIT. The method evaluates the empirical PGF at damped query points that stay inside the unit disk while preserving the exact sum-of-exponentials structure, recovers multidimensional rate vectors through a global subspace and a non-unitary similarity diagonalization, estimates rates by nonnegative least squares and weights by simplex-constrained least squares, and finishes with a quasi-Newton refinement of the exact PGF loss using a closed-form analytic gradient. Because the sketch has a fixed size, the recovery cost and storage do not grow with the number of observations, which we confirm empirically against full-data EM. The final version will complete the empirical study by separating algorithmic comparisons under the finite product-Poisson mixture model from model-level comparisons involving alternative count-data models.

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